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Digital Transformation

DATA COLLECT

Methodology for data reading, extraction and storage.

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Aim of the document

Describe the connection details used for BLE connection and telemetry data extraction from ImberaLink controllers with firmware 1.26, 1.36 and 1.41

Firmware versions

We have three different versions to collect:

1. **1.26** version, which include the following versions:
    a. 121
    b. 122
    c. 123
    d. 126
2. **1.36** version, which include the following versions:
    a. 136
    b. 137
    c. 141
    d. 143
    e. 170
    f. 210
    g. 230
    h. 240
    i. 241
    j. 242
3. **6.0** version, which include the following versions:
    a. 619
    b. 620
    c. 621
    d. ...
    e. 635

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Firmware 1.26

Steps to obtain the measurement buffer (MEASUREMENTS):

1. Connect to PGED service
00040000-0000-1000-8000-00805f9baaaa
2. Read HEADER
00040001-0000-1000-8000-00805f9baaaa

The response bring us the number and size of samples, in case it has no samples we go on to read the buffer of events.

3. The **measurements** buffer is requested via the following feature: This step is repeated until the cooler no longer responds with further measurements.

Description	ID
PGED_WRITE	00040002-0000-1000-8000-00805f9baaaa

The instruction sent is as follows:

0001**0003**00000000**0001**

Instruction sent breakdown:

0001 charge

0003 measurements buffer

00000000 Index zero

0001 one element

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4. Read the answer through the following feature:

Description	ID
PGED_READ	00040003-0000-1000-8000-00805f9baaaa

Send data.

5. Delete each of the measurement elements sent through the following feature:

Description	ID
PGED_WRITE	00040002-0000-1000-8000-00805f9baaaa

Deleting instruction

00020003000000000001

Instruction breakdown

0002 Delete

0003 Measurements

00000000 Index zero

0001 one element

6. **Event** reading process started (EVENTS)

HEADER update is requested in the following feature:

Description	ID
PGED_WRITE	00040002-0000-1000-8000-00805f9baaaa

Instruction sent

0003

Instruction breakdown

0003 header update

Sample number and size are extracted in the answer.

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- EVENT buffer is requested trough the following feature:

This step is repeated until the cooler responds 00000000 or until the EVENTS buffer is empty and the connection is closed.

Description	ID
PGED_WRITE	00040002-0000-1000-8000-00805f9baaaa

The instruction sent is as follows:

00010001000000000001

Instruction breakdown

0001 charge

0001 Event buffer

00000000 Index zero

0001 one element

7. Read the answer through the following feature:

Description	ID
PGED_READ	00040003-0000-1000-8000-00805f9baaaa

Send data.

8. Delete each of the event items sent throug the following feature:

Description	ID
PGED_WRITE	00040002-0000-1000-8000-00805f9baaaa

Deleting instruction

00020001000000000001

Instruction breakdown

0002 Delete

0001 Events

00000000 Index zero

0001 one element

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Firmware 1.36 / 2.42

Steps to obtain the events buffer (EVENTS).

1. Connect with the service: 00070000-0000-1000-8000-00805f9baaaa
2. Read the header: 00070001-0000-1000-8000-00805f9baaaa

The response bring us the number and size of samples, in case it has no samples we go on to read the buffer of events (step 5).

3. The **event** buffer is requested through the following feature:
This step is repeated until the cooler no longer responds with any more events

Description	ID
PGED_WRITE	00070002-0000-1000-8000-00805f9baaaa

The instruction sent is as follows:
00018006000000000XXXX
Instruction sent breakdown:
0001 charge
8006 Event buffer
00000000 Index zero
XXXX number of element

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4. Delete each of the events elements sent through the following feature:

Description	ID
PGED_WRITE	00070002-0000-1000-8000-00805f9baaaa

Deleting instruction:
00028006000000000001
Instruction breakdown:
0002 Delete
8006 Event
00000000 index zero
0001 number of elements to be deleted

5. The **measurements** buffer is requested through the following feature:
This step is repeated until the cooler no longer responds with any more measurements.

Description	ID
PGED_WRITE	00070002-0000-1000-8000-00805f9baaa

The instruction sent is as follows:
00018008000000000XXXX
Instruction breakdown:
0001 charge
8008 measurements buffer
00000000 Index zero
XXXX number of elements
When the notification arrives the measurements are sent.
6. Delete each of the measurement elements sent through the following feature:

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Description	ID
PGED_WRITE	00070002-0000-1000-8000-00805f9baaaa

Deleting instruction:

0002800800000000XXXX

Instruction breakdown:

0002 Delete

8008 Measurement

00000000 Index zero

XXXX number of elements to be deleted

Firmware 6.0

Steps to obtain the events buffer (EVENTS).

1. Connect with the service: 00070000-0000-1000-8000-00805f9baaaa
2. Read the header: 00070001-0000-1000-8000-00805f9baaaa

The response bring us the number and size of samples, in case it has no samples we go on to read the buffer of measurements (step 5).

3. The **event** buffer is requested through the following feature:
This step is repeated until the cooler no longer responds with any more events

Description	ID
PGED_WRITE	00070002-0000-1000-8000-00805f9baaaa

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The instruction sent is as follows:

00018015000000000XXXX

Instruction sent breakdown:

0001 charge

8015 Event buffer

00000000 Index zero

XXXX number of element

When the notification arrives, the events are sent to the Cloud with the model defined for exchange.

4. Delete each of the events elements sent through the following feature:

Description	ID
PGED_WRITE	00070002-0000-1000-8000-00805f9baaa

Deleting instruction:

00028015000000000XXXX

Instruction breakdown:

0002 Delete

8015 Event

00000000 index zero

XXXX number of elements to be deleted

5. The **measurements** buffer is requested through the following feature:
This step is repeated until the cooler no longer responds with any more measurements.

Description	ID
PGED_WRITE	00070002-0000-1000-8000-00805f9baaa

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The instruction sent is as follows:

00018017000000000XXXX

Instruction breakdown:

0001 charge

8017 measurements buffer

00000000 Index zero

XXXX number of elements

When the notification arrives the measurements are sent.

6. Delete each of the measurement elements sent through the following feature:

Description	ID
PGED_WRITE	00070002-0000-1000-8000-00805f9baaaa

Deleting instruction:

00028017000000000XXXX

Instruction breakdown:

0002 Delete

8017 Measurement

00000000 Index zero

XXXX number of elements to be deleted

7. The **alarms** buffer is requested through the following feature:

This step is repeated until the cooler no longer responds with any more alarms.

Description	ID
PGED_WRITE	00070002-0000-1000-8000-00805f9baaaa

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The instruction sent is as follows:

00018016000000000XXXX

Instruction breakdown:

0001 charge

8016 alarms buffer

00000000 Index zero

XXXX number of elements

When the notification arrives the alarms are sent.

8. Delete each of the alarms elements sent through the following feature:

Description	ID
PGED_WRITE	00070002-0000-1000-8000-00805f9baaaa

Deleting instruction:

00028016000000000XXXX

Instruction breakdown:

0002 Delete

8016 Alarms

00000000 Index zero

XXXX number of elements to be deleted

9. The **temporal measures** buffer is requested through the following feature:

This step is repeated until the cooler no longer responds with any more temporal measures.

Description	ID
PGED_WRITE	00070002-0000-1000-8000-00805f9baaaa

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The instruction sent is as follows:

00018018000000000XXXX

Instruction breakdown:

0001 charge

8018 temporal measurements buffer

00000000 Index zero

XXXX number of elements

When the notification arrives the temporal measurements are sent.

10. Delete each of the temporal measurements elements sent through the following feature:

Description	ID
PGED_WRITE	00070002-0000-1000-8000-00805f9baaaa

Deleting instruction:

00028018000000000XXXX

Instruction breakdown:

0002 Delete

8018 Temporal measurements

00000000 Index zero

XXXX number of elements to be deleted

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Splitting trama

- **Firmware 1.26**
  - Events (0001): 6 bytes
  - Measurements (0003): 15 bytes
- **Firmware 1.36 - 2.42**
  - Events (8006): 11 bytes
  - Measurements (8008): 9 bytes
- **Firmware 6.0 - 6.35**
  - Events (8015): 15 bytes
  - Measurements (8017): 12 bytes
  - Temporal measurements (8018): 34 bytes
  - Alarms (8016): 15 bytes

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VAULT Service

Only for version 1.4.9 from CIL2 controller

This part describes how to send VAULT commmand through Bluetooth to CIL controllers.

1. Connect with the service with UUID: **000a0001-0000-1000-8000-00805f9baaaa**
2. Send the command to activate VAULT mode: **A013A17EFE9E7A3F69EC4A4A53B749CD7A60C2**
3. Send the command to validate VAULT status: **A013A17EFE9E7A3F69EC4A4A53B749CD7A60C4**
4. When VAULT is active the response is going to be: **A3040101** and **Lo** will be displayed on the controller's display. If VAULT mode is not active, the response is going to be **A3040100** and the controller will restart.
5. Send the command to exit VAULT mode: **A013A17EFE9E7A3F69EC4A4A53B749CD7A60C3**

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